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Support unit, transfer device, and image forming apparatus

Abstract

A support unit includes: a first support to support and position one end of a first shaft of a pressure unit, including a pressure rotator to apply pressure to an object, in each of: an assembly direction in which the pressure unit is assembled to the support unit; and an orthogonal direction orthogonal to each of the assembly direction and an axial direction of the first shaft; a second support having a groove supporting another end of the first shaft; and a third support supporting a second shaft of the pressure rotator. The first support, the second support, and the third support the pressure unit rotatable around the first shaft. The groove of the second support has an arc shape having a center on an axial center of the second shaft; and an opening at an entrance of the groove in the assembly direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This patent application is based on and claims priority pursuant to 35 U.S.C. § 119(a) to Japanese Patent Application No. 2023-084195, filed on May 22, 2023, in the Japan Patent Office, the entire disclosure of which is hereby incorporated by reference herein.

BACKGROUND

Technical Field

(2) Embodiments of the present disclosure relate to a support unit, a transfer device, and an image forming apparatus.

(3) A support unit is known that allows a pressure unit to rotate freely around the reference shaft as a pivot point. The support unit includes a first support, a second support, and a third support. The first support supports one end of the reference shaft used for positioning the pressure unit that presses a pressed member via a pressing member. The second support supports the other end of the reference shaft. The third support supports the shaft of the pressing member.

(4) A typical support unit rotatably supports a secondary transfer unit serving as a pressure unit.

(5) However, the pressure unit may twist when the pressing member applies pressure to the pressed member.

SUMMARY

(6) An embodiment of the present disclosure provides a support unit including: a first support to support and position one end of a first shaft of a pressure unit, including a pressure rotator to apply pressure to an object, in each of: an assembly direction in which the pressure unit is assembled to the support unit; and an orthogonal direction orthogonal to each of the assembly direction and an axial direction of the first shaft; a second support having a groove supporting another end of the first shaft; and a third support supporting a second shaft of the pressure rotator. The first support, the second support, and the third support the pressure unit rotatable around the first shaft. The groove of the second support has an arc shape having a center on an axial center of the second shaft; and an opening at an entrance of the groove in the assembly direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A more complete appreciation of embodiments of the present disclosure and many of the attendant advantages and features thereof can be readily obtained and understood from the following detailed description with reference to the accompanying drawings, wherein:

(2) FIG. 1 is a schematic diagram of an image forming apparatus according to an embodiment of the present disclosure;

(3) FIGS. 2A, 2B, and 2C are diagrams each illustrating a configuration of a secondary transfer device;

(4) FIG. 3 is a diagram illustrating inclination adjustment;

(5) FIGS. 4A-1, 4A-2, 4B-1, and 4B-2 are diagrams illustrating how the secondary transfer unit in FIGS. 2A, 2B, and 2C that is twisted under normal conditions is attached to a drawer housing;

(6) FIGS. 5A-1, 5A-2, 5B-1, and 5B-2 are diagrams illustrating how a secondary transfer roller of a secondary transfer unit that is twisted under normal conditions contacts an intermediate transfer belt;

(7) FIGS. 6A-1, 6A-2, 6B-1, and 6B-2 are diagrams each illustrating how a secondary transfer roller contacts an intermediate transfer belt when the secondary transfer unit is twisted under normal conditions and a rear portion of the secondary transfer roller is positioned higher than a front portion of the secondary transfer roller, according to an embodiment of the present disclosure;

(8) FIG. 7 is an enlarged view of a part of a rotary shaft on the rear side;

(9) FIG. 8 is a diagram of a rotary shaft on the rear side according to a first modification of an embodiment of the present disclosure;

(10) FIGS. 9A and 9B are diagrams each illustrating a rotary shaft on the rear side according to a second modification of an embodiment of the present disclosure;

(11) FIG. 10 is a diagram of a rotary shaft on the rear side according to a third modification of an embodiment of the present disclosure;

(12) FIGS. 11A, 11B, and 11C are diagrams each illustrating a configuration of the rotary shaft on the rear side according to the third modification; and

(13) FIG. 12 is a diagram of a rotary shaft on the right side according to a fourth modification of an embodiment of the present disclosure.

(14) The accompanying drawings are intended to depict embodiments of the present disclosure and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. Also, identical or similar reference numerals designate identical or similar components throughout the several views.

DETAILED DESCRIPTION

(15) In describing embodiments illustrated in the drawings, specific terminology is employed for the sake of clarity. However, the disclosure of this specification is not intended to be limited to the specific terminology so selected and it is to be understood that each specific element includes all technical equivalents that have a similar function, operate in a similar manner, and achieve a similar result.

(16) Referring now to the drawings, embodiments of the present disclosure are described below. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

(17) According to one aspect of the present disclosure, the occurrence of twisting in the pressure unit is reduced, and the pressure unit can be easily assembled to the support unit. The following describes a developing device, a process cartridge, and an image forming apparatus, according to an embodiment of the present disclosure, with reference to the accompanying drawings. It is to be understood that those skilled in the art can easily modify and change the present disclosure within the scope of the appended claims to form other embodiments, and these modifications and changes are included in the scope of the appended claims. The following describes an illustrative embodiment and does not limit the present disclosure.

(18) With reference to drawings, the following describes an embodiment of the present disclosure. FIG. 1 is a schematic view of a configuration of an image forming apparatus, or a printer **100**. In the present embodiment, the image forming apparatus is an electrophotographic color printer. The printer **100** is provided with four image forming units **1Y**, **1M**, **1C**, and **1K** that form toner images of yellow (Y), magenta (M), cyan (C), and black (K), respectively. The printer **100** is also provided with, for example, a transfer unit **30** as an intermediate transfer device, a secondary transfer unit **40**, a cassette **60** that stores a recording material P as a to-be-conveyed object, and a fixing device **90**.

(19) The four image forming units **1Y**, **1M**, **1C**, and **1K** form an image forming section and use toners of Y, M, C, and K that are different color toners as powder developers. The image forming units **1Y**, **1M**, **1C**, and **1K** have a similar structure except the color of toner. The image forming units **1Y**, **1M**, **1C**, and **1K** include drum-shaped photoconductors **2Y**, **2M**, **2C**, and **2K** serving as image bearers, photoconductor cleaners **3Y**, **3M**, **3C**, and **3K**, dischargers, charging devices **6Y**, **6M**, **6C**, and **6K**, developing devices **8Y**, **8M**, **8C**, and **8K**, respectively.

(20) The photoconductors **2Y**, **2M**, **2C**, and **2K** have surfaces that are evenly charged by the charging devices **6Y**, **6M**, **6C**, and **6K**. Subsequently, the surfaces of the photoconductors **2Y**, **2M**, **2C**, and **2K** are optically scanned by exposure light such as the laser beams emitted from a plurality of optical writing units **101** disposed above the image forming units **1Y**, **1M**, **1C**, and **1K**. As a result, multicolor electrostatic latent images are formed. The developing devices **8Y**, **8M**, **8C**, and **8K** develop the electrostatic latent images on the photoconductors **2Y**, **2M**, **2C**, and **2K** with yellow, magenta, cyan, and black toners, into visible toner images T, respectively. Thus, the toner images T are formed on the photoconductors **2Y**, **2M**, **2C**, and **2K**. The toner images T on the photoconductors **2Y**, **2M**, **2C**, and **2K** are primarily transferred and borne onto the front surface of the intermediate transfer belt **31** composed of an endless belt.

(21) The intermediate transfer unit **30** as an intermediate transfer device is disposed below the image forming units **1Y**, **1M**, **1C**, and **1K**. The intermediate transfer unit **30** stretches and rotates the intermediate transfer belt **31** in the clockwise direction in FIG. 1. In the present embodiment, a direction of rotation of the intermediate transfer belt **31** is referred to as a “belt travel direction a”

indicated by an arrow a in FIG. 1.

(22) In addition to the intermediate transfer belt **31**, the transfer unit **30** includes a drive roller **32**, a secondary-transfer backside roller **33**, a cleaning backup roller **34**, four primary transfer rollers **35Y**, **35M**, **35C**, and **35K**, and a pre-transfer roller **37**. The intermediate transfer belt **31** is looped around and stretched taut between the drive roller **32**, the secondary-transfer backside roller **33**, the cleaning backup roller **34**, the four primary transfer rollers **35Y**, **35M**, **35C**, and **35K**, and the pre-transfer rollers **37**. As a driver such as a drive motor drives and rotates the drive roller **32** clockwise, the intermediate transfer belt **31** rotates clockwise.

(23) A secondary transfer unit **40** as a pressure unit that includes a secondary transfer belt **102** is arranged outside and below the loop of the intermediate transfer belt **31**. The secondary transfer belt **102** according to the present embodiment serves as a belt member or a transfer belt. The secondary transfer belt **102** is looped around multiple tension rollers including the secondary transfer roller **41** as a pressure member or pressure rotator.

(24) A cassette **60** that serves as a container and stores a plurality of recording materials P in a stacked state is arranged below the secondary transfer unit **40**. In the cassette **60**, a roller **60a** contacts an uppermost recording material P of the bundle of recording media and rotates at a predetermined timing to feed the recording material P from the cassette **60** to a conveyance path **65** toward a secondary transfer nip N2. The recording material P that is fed and sent to the conveyance path **65** is then sent by a registration roller pair **61** to the secondary transfer nip N2 at a synchronized timing in the secondary transfer nip N2 with the toner image on the front surface of the intermediate transfer belt **31**.

(25) In the secondary transfer nip N2, the toner image on the outer circumferential surface of the intermediate transfer belt **31** is collectively transferred onto the recording material P by a secondary transfer electric field and a nip pressure applied thereto, thereby forming a full-color toner image in combination with white color of the recording material P.

(26) The fixing device **90** is disposed downstream from the secondary transfer nip N2 in a conveyance direction b of the recording medium P. The fixing device **90** includes a heating roller **91** as a heating rotator, a fixing roller **93**, a support roller **96**, and a tension roller **95** serving as a tensioner that are support rollers for the fixing belt **94**.

(27) The fixing device **90** also includes a pressure roller **92** as a pressing rotator that contacts the fixing roller **93**. The fixing belt **94** is sandwiched by the pressure roller **92** and the fixing roller **93**. The recording material P to which the toner image has been transferred is fed to the fixing device **90**. The recording material P is nipped at a fixing nip at which the fixing roller **93** and the pressure roller **92** are in contact with each other via the fixing belt **94**. In the fixing device **90**, the heating roller **91** includes a heat source therein. Heat of the heat source transfers from the heating roller **91** to the recording material P via the fixing belt **94** and softens toner in the full-color toner image at the fixing nip. The heat and pressure in the fixing nip fixes the full-color toner image onto the recording material P. After the toner image is fixed on the recording material P, the recording material P is ejected from the fixing device **90**, outside the printer **100**.

(28) As illustrated in FIG. 1, the secondary transfer device **400** according to the present embodiment has the secondary transfer unit **40** as a belt unit or a pressure unit, and a drawer housing **113** as a support unit to support the secondary transfer unit **40** in a rotatable manner.

(29) The secondary transfer unit **40** according to the present embodiment is provided with five tension rollers that support the secondary transfer belt **102** on its inner surface. More specifically, the five tension rollers include the secondary transfer roller **41**, a separation roller **42**, a driven roller **43**, a drive roller **44**, and an imbalance correction roller **45**. also includes a tension roller **46** that presses the secondary transfer belt **102** from its outer circumferential surface. The secondary transfer unit **40** is supported by the drawer housing **113**. The drawer housing **113** is a unit support housing that can be pulled out toward the front of the device body, or in the X-direction. FIG. 1 is a schematic front view of the printer **100**, and the X-direction, the Y-direction, and the Z-direction

illustrated in FIG. 1 are defined as follows. The X-direction is one of the front-to-back direction of the printer **100**, specifically toward the front. The Y-direction corresponds to the left-to-right direction of the printer **100**, specifically toward the right. The Z-direction is the vertical direction, which is the direction (or the direction of assembly) in which the drawer housing **113** of the secondary transfer unit is assembled.

(30) FIG. 2 is a schematic diagram illustrating a configuration of the secondary transfer device **400**. The X-direction, Y-direction, and Z-direction in FIGS. 2A, 2B, and 2C are the same as in FIG. 1. FIG. 2B is a side view of the secondary transfer unit **40** in a direction from the left of the printer. FIG. 2A is a rear view of the secondary transfer unit **40**, taken from the rear side of the printer. FIG. 2C is a front view of the secondary transfer unit **40**, taken from the front side of the printer.

(31) The secondary transfer unit **40** according to the present embodiment includes a pair of roller support plates **103** as a roller support that supports multiple tension rollers around which the secondary transfer belt **102** is looped. The secondary transfer unit **40** is supported by the drawer housing **113**. The drawer housing **113** is a support unit that can be pulled out in the X-direction of the device body.

(32) The secondary transfer roller **41** according to the present embodiment, which forms the secondary transfer nip N2 (see FIG. 1) with the intermediate transfer belt **31**, freely rotates around a secondary transfer roller shaft **106**, which is a fixed shaft. The secondary transfer roller shaft **106** is supported by contact surfaces **110** and **111** as a third support of rotary arms **108** and **109** (or a pressing body or pressing bodies), and is pressed against the intermediate transfer belt **31** by pressing forces from the rotary arms **108** and **109**. The contact surfaces **110** and **111** are substantially perpendicular to a vector connecting the centers of the secondary transfer roller **41** and the secondary-transfer backside roller **33** opposed to the secondary transfer roller **41** with the intermediate transfer belt **31** interposed therebetween. The contact surfaces **110** and **111** serve as sub-reference planes for positioning the secondary transfer unit **40**.

(33) The rotary arms **108** and **109** are supported by rotary shafts **116** and **117**, which are support portions integral with the drawer housing **113**, so as to be rotatable about the center shaft. Springs **114** and **115** between the rotary arms **108** and **109** and the drawer housing **113** rotationally urge the rotary arms **108** and **109**. The rotary arms **108** and **109** press the secondary transfer unit **40** against the intermediate transfer belt **31** via the contact surfaces **110** and **111**.

(34) The rotary shaft **116** as a first support on the front side of the secondary transfer unit **40** has a hole **116a** that supports the front end of a shaft **118** as a reference shaft in the X-direction. The shaft **118** serves as a primary reference for positioning the secondary transfer unit **40**. The front portion of the rotary shaft **116**, which is one end in the axial direction, supports the shaft **118** and the front portion of the rotary arm **108** such that the rotation center of the shaft **118** matches the rotation center of the front portion of the rotary arm **108**. The rotary shaft **116** on the front side swingably supports the front end portion of the shaft **118** of the secondary transfer unit **40**. The shaft **118** of the secondary transfer unit **40** is an irrotational axis whose both ends are supported by the pair of roller support plates **103**, and rotatably supports the driven roller **43** illustrated in FIG. 1 between the pair of roller support plates **103**.

(35) The rotary shaft **117** as a second support has a groove **117a** that engages and supports the rear end portion of the shaft **118** of the secondary transfer unit **40**. A pair of side surfaces of the groove **117a** are shaped in a radius (R) that is concentric with the center of the secondary transfer roller **41**. The groove **117a** is shaped as an arc, centered on the axis of the secondary transfer roller **41**. The rear end portion of the shaft **118** is supported in the groove **117a**, allowing the shaft **118** to move in a manner concentric with the center of the secondary transfer roller **41**.

(36) The secondary transfer unit **40** is supported by the drawer housing **113** via the rotary shafts **116** and **117** and the contact surfaces **110** and **111** of the rotary arms **108** and **109**. The secondary transfer unit **40** is detachably attachable to the drawer housing **113** in the Z-direction or the vertical direction. As the drawer housing **113** is drawn from the body of the image forming apparatus, the

secondary transfer unit **40** can be detachably attached to the drawer housing **113** in the Z-direction.

(37) The rotary arm **108** on the front side and the rotary arm **109** on the rear side are rotationally driven by separate pressure drive devices, and the pressure of the rotary arms **108** and **109** to the secondary transfer unit **40** can be adjusted by the respective pressure drive devices.

(38) In the present embodiment, various types of recording materials P may be used such as plain paper, thick paper, a postcard, an envelope, thin paper, coated paper, art paper, tracing paper, and a transparency for overhead projector (OHP). The optimum secondary-transfer pressure or the optimum pressure at the transfer nip varies depending on the type of the recording material P. In order to handle such a situation, in the present embodiment, the secondary-transfer pressure is adjusted depending on the type of the recording material P. For example, information about a type of the recording material P placed in the cassette **60** is input by a user operating an operation panel included in the printer body. The controller of the printer sets the drive time of the pressure motors in the respective pressure drive devices based on the input information about the type of recording materials P. When the pressure motors are stepping motors, the controller sets the number of steps based on the input information about the type of recording material P.

(39) As the pressure motors drives for the set drive time or the set number of steps, the pressing force applied from the rotary arms **108** and **109** to the secondary transfer roller shaft **106** is adjusted according to the type of sheet or the recording material P. Due to such a configuration, the pressing force that is applied to the secondary transfer nip N2 is adjusted to the pressing force suitable for the type of sheet to achieve secondary-transfer pressure suitable for the type of sheet.

(40) In the present embodiment, the following advantages are obtained by rotatably attaching the rotary arms **108** and **109** to the rotary shafts **116** and **117**, respectively, which supports the secondary transfer unit **40**. In other words, the rotation center of the secondary transfer unit and the rotation center of the rotary arm can be substantially aligned with each other, and the change in the contact position of the rotary arm with the secondary transfer roller shaft **106** can be reduced by adjusting the pressing force.

(41) This minimizes the disturbance in the relation between the pressing force and the rotational position of the rotary arm, thus facilitating the adjustment of the pressing force.

(42) In particular, the front side of the shaft **118** is inserted into the hole **116a** centered on the axis of the rotary shaft **116**. This positions the secondary transfer unit **40** in the Z-direction and the Y-direction. The rotation center of the secondary transfer unit **40** and the rotation center of the rotary arm **108** coincide with each other. This minimizes, for the front side, the disturbance in the relation between the pressing force and the rotational position of the rotary arm, thus facilitating the adjustment of the pressing force.

(43) In this embodiment, the front end portion (one end in the axial direction) of the shaft **118** of the secondary transfer unit **40** is swingably supported, and the rear end portion (the other end) of the shaft **118** of the secondary transfer unit **40** is inserted into the groove **117a** and is supported movably concentrically with the center of the secondary transfer roller **41**. Thus, the secondary transfer unit **40** is supported by the drawer housing **113** without its orientation in the Y-direction being restricted. As a result, the secondary transfer unit **40** can be supported in a position inclined in the Z-direction with respect to the X-direction within the apparatus.

(44) The secondary transfer unit **40** may be twisted when it is in the standalone belt unit state (referred to as “natural condition”) before installation into the unit support housing, due to manufacturing errors or similar factors. In this case, the orientation of the roller support plate **103** on the front side in the X-direction differs from the orientation of the roller support plate **103** on the rear side in the X-direction. As a result, the multiple rollers around which the secondary transfer belt **102** supported by the roller support plate **103** is looped are inclined with respect to the X-direction, and the secondary transfer belt **102** may be skewed to one side in the X-direction while the secondary transfer belt **102** is rotating. Such a skew of the belt to one side may be referred to as belt imbalance in the following description.

(45) In view of this, in the present embodiment, the inclination of one of the rollers over which the secondary transfer belt **102** is stretched with respect to the X-direction can be adjusted. Adjusting the inclination of the roller with respect to the X-direction reduces the imbalance, or skew of the secondary transfer belt **102** due to the twist of the secondary transfer unit **40** under normal conditions.

(46) FIG. **3** is a diagram illustrating the inclination adjustment.

(47) In the present embodiment, the tension roller **46** is capable of inclination adjustment. The front portion of the tension roller **46** is supported by a front portion of the roller support plate **103** through an inclination adjuster **202**. A roller other than the tension roller **46** may be a roller whose inclination can be adjusted.

(48) The inclination adjuster **202** is rotatably supported by the pin **203** that is arranged on the front portion of the roller support plate **103**. The inclination adjuster **202** according to the present embodiment is provided with a round screw insertion slot **205** around the pin **203**, i.e., the rotation center of the inclination adjuster **202**.

(49) As the inclination adjuster **202** is driven to rotate around the pin **203**, the front side of the tension roller **46** moves in the direction of the arrow A in the drawing, and the inclination of the tension roller **46** with respect to the X-direction (axial direction) is adjusted. For example, when the secondary transfer belt **102** is skewed to the rear side due to the torsion of the secondary transfer unit **40** under normal conditions, the inclination is adjusted as follows. The inclination of the tension roller **46** with respect to the X-direction is adjusted at the portion where the secondary transfer belt **102** is looped over the tension roller **46** such that the secondary transfer belt **102** is skewed to the front side. After the inclination adjustment is complete, the inclination adjuster **202** is fastened to a front portion of the roller support plate **103** using a screw **204**. Due to such a configuration, the belt imbalance or skew of the secondary transfer belt **102** due to the twist of the secondary transfer unit **40** under normal conditions is reduced.

(50) When the rotary shafts **116** and **117** support the shaft **118** in a manner that the shaft **118** remains fixed (or cannot move) within the YZ plane, and the secondary transfer unit **40** is mounted in the drawer housing **113** without restricting its orientation around the X-axis, the following troubles may arise. In other words, when the secondary transfer unit **40** twisted under natural conditions is brought into contact with the intermediate transfer belt **31** through the secondary transfer belt **102** by the rotary arms **108** and **109**, twisting occurs in the secondary transfer roller **41**. As a result, the optimum inclination relationship among the rollers, which is maintained by the inclination adjuster **202** to prevent excessive belt skew, is disrupted, leading to an increased skew of the secondary transfer belt **102**. As a result, the speed of the skew of the secondary transfer belt **102** exceeds acceptable limits, causing the guides at both ends of the secondary transfer belt **102** to strongly collide with the restricting member of the imbalance correction roller **45**. This frequent contact can lead to premature wear and potentially damage the secondary transfer belt **102**.

(51) However, in the present embodiment, with the shaft **118** supported and the secondary transfer unit **40** attached to the drawer housing **113**, the orientation of the secondary transfer unit **40** around the Y-direction is maintained without restriction. In other words, the front end portion of the shaft **118** is pivotally supported by the rotary shaft **116** on the front side, and the rear end portion of the shaft **118** is supported by the rotary shaft **117** on the rear side in such a manner that the shaft **118** can move around the secondary transfer roller shaft. This configuration allows the secondary transfer unit **40** to be supported without any restriction on its orientation around the Y-axis. With such a configuration, the secondary transfer unit **40** that is inclined in the Z-direction with respect to the X-direction can be supported by the drawer housing **113**. Thus, by using the rotary arms **108** and **109**, the twisting of the secondary transfer roller **41** when the secondary transfer roller **41** contacts the intermediate transfer belt **31** is reduced, which helps reduce the skew of the secondary transfer belt **31**. This also prevents the skew from worsening and keeps the speed of the skew within acceptable limits. This is described below with reference to the drawings.

(52) FIGS. 4A-1, 4A-2, 4B-1, and 4B-2 are diagrams illustrating how the secondary transfer unit **40** that is twisted under normal conditions is attached to the drawer housing **113**.

(53) In FIGS. 4A-1, 4A-2, 4B-1, and 4B-2, the solid lines indicate the front side, and the broken lines indicates the rear side.

(54) For example, as illustrated in FIGS. 5A-1, 5A-2, 5B-1, and 5B-2, the twisting of the secondary transfer unit **40** causes a misalignment between the front portion and the rear portion of the shaft **118**. As it stands, with the front portion of the shaft **118** supported by the rotary shaft **116**, the rear portion of the shaft **118** cannot be inserted into the groove **117a** of the rotary shaft **117**.

(55) In the present embodiment, by swingably supporting the front portion of the shaft **118** with respect to the rotary shaft **116**, as illustrated in FIG. 4B-1, the secondary transfer unit **40** can be attached to the drawer housing **113** while being inclined in the Y-direction with respect to the X-direction. By inclining the secondary transfer unit **40** in the Y-direction relative to the X-direction, the rear portion of the shaft **118** can be inserted into the groove **117a** of the rotary shaft **117**, allowing the twisted secondary transfer unit **40** to be supported by the drawer housing **113**.

(56) FIGS. 5A-1, 5A-2, 5B-1, and 5B-2 are diagrams illustrating how the secondary transfer roller **41** of the secondary transfer unit **40** that is twisted under normal conditions contacts the intermediate transfer belt **31**.

(57) As illustrated in FIGS. 5A-1 and 5A-2, due to the twist of the secondary transfer unit **40**, the rear portion of the secondary transfer roller **41** indicated by the broken line in the drawing may be positioned lower than its front portion indicated by the solid line in the drawing. In such cases, the front portion of the secondary transfer roller **41** contacts the intermediate transfer belt **31** earlier than the rear portion of the secondary transfer roller **41**.

(58) As described above, according to the present embodiment, the rotary arm **108** on the front side and the rotary arm **109** on the rear side are driven to rotate independently by different pressure drivers. In this configuration, when the front portion of the secondary transfer roller **41** contacts the intermediate transfer belt **31**, the front portion of the rotary arm **108** stops rotating, and only the rear portion of the rotary arm **109** continues rotating.

(59) In view of these circumstances, when the orientation of the secondary transfer unit **40** in the Y-direction is restricted and the secondary transfer unit **40** is supported by the drawer housing **113**, the rear portion of the secondary transfer roller **41** rotates on the shaft **118** in the X-direction. As a result, the twisting of the secondary transfer unit **40** changes, the relationship between the inclinations of the rollers gets worse, the belt skew is deteriorated, and the speed of the skew of the belt may exceed the allowable value. As a result, the speed of the skew of the secondary transfer belt **102** exceeds acceptable limits, causing the guides at both ends of the secondary transfer belt **102** to strongly collide with the restricting member of the imbalance correction roller **45**. This frequent contact can lead to premature wear and potentially damage the secondary transfer belt **102**.

(60) In contrast, in the present embodiment, the shaft **118** on the front side is supported so as to be swingable, and the shaft **118** on the rear side is supported so as to be movable concentrically with the rotation center of the secondary transfer roller **41**. Thus, the secondary transfer unit **40** is supported by the drawer housing **113** without its orientation in the Y-direction being restricted. In this configuration, after the front portion of the secondary transfer roller **41** contacts the intermediate transfer belt **31**, the orientation of the secondary transfer unit **40** around the Y-direction changes in such a manner that the entire rear portion of the secondary transfer unit **40** is lifted by the pressing force of the rotary arm **109** on the rear side as indicated by arrow A1 in FIG. 5A-1. As a result, as illustrated in FIG. 5B-2, the secondary transfer unit **40** is inclined in the Z-direction with respect to the X-direction, and the entire region of the secondary transfer roller **41** in the X-direction is brought into contact with the intermediate transfer belt **31**.

(61) In this way, the secondary transfer unit **40** is only inclined in the Z-direction with respect to the X-direction, and the twist of the secondary transfer unit **40** is hardly changed from the natural

state. Thus, the inclination of each roller can be maintained in an optimum relationship for reducing the belt skew, the worsening of the belt skew can be prevented, and the belt skew speed can be maintained at an allowable value or less.

(62) FIGS. 6A-1, 6A-2, 6B-1, and 6B-2 are diagrams illustrating cases where a rear portion of the secondary transfer roller **41** is positioned higher than a front portion of the secondary transfer roller **41** when the secondary transfer unit **40** is twisted under normal conditions, according to the present embodiment.

(63) As illustrated in FIG. 6A-1 and FIG. 6A-2, when a rear portion of the secondary transfer roller **41** is positioned higher than a front portion of the secondary transfer roller **41** when the secondary transfer unit **40** is twisted under normal conditions, such a rear portion of the secondary transfer roller **41** indicated by broken lines in FIG. 6A-1 contacts the intermediate transfer belt **31** earlier than the front portion of the secondary transfer roller **41** indicated by a solid line in FIG. 6A-1.

(64) After the rear portion of the secondary transfer roller **41** contacts the intermediate transfer belt **31**, when the rotation of the rotary arm **108** on the front side continues to cause the front portion of the secondary transfer roller **41** to contact the intermediate transfer belt **31**, the front portion of the secondary transfer roller **41** rotates counterclockwise around the shaft **118** as a pivot point, as illustrated in FIG. 6A-1. At this time, the secondary transfer unit **40** is rotated counterclockwise as in FIG. 6A-2 by the reaction from the intermediate transfer belt **31** (or the secondary-transfer backside roller **33**), that is, the secondary transfer unit **40** changes its orientation around the Y-direction. As a result, as illustrated in FIG. 6B-1 and FIG. 6B-2, when the front portion of the secondary transfer roller **41** contacts the intermediate transfer belt **31**, the secondary transfer unit **40** is inclined with respect to the drawer housing **113** such that the rear portion of the secondary transfer unit **40** is positioned lower than the front portion of the secondary transfer unit **40**.

(65) As described above, even when the secondary transfer roller **41** is twisted such that the rear portion thereof is positioned higher than the front portion thereof, the orientation of the secondary transfer unit **40** around the Y-direction with respect to the drawer housing **113** is changed, and the entire secondary transfer roller **41** contacts the intermediate transfer belt **31**. Thus, the inclination of each roller can be maintained in an optimum relationship for reducing the belt skew, the worsening of the belt skew can be prevented.

(66) Further, as the groove **117a** is formed in an arc shape around the rotation center of the secondary transfer roller **41** in the present embodiment, the positional displacement between the front portion and the rear portion of the secondary transfer roller **41** in the Y-axis direction when the secondary transfer roller **41** contacts the intermediate transfer belt **31** can be reduced.

Accordingly, the changes in the pressure at the transfer nip can be reduced.

(67) FIG. 7 is an enlarged view of a part of the rotary shaft **117** on the rear side as a second support.

(68) As illustrated in FIG. 7, the groove **117a** has a groove shape such that a shaft insertion port **119** that is an opening of the groove **117a** into which the rear end portion of the shaft **118** is inserted when the secondary transfer unit **40** is assembled to the drawer housing **113** and is a reference-shaft insertion port, can be observed from above. Thus, the rear end portion of the shaft **118** can be inserted into the groove **117a** merely by moving the secondary transfer unit **40** vertically downward, which is the assembly direction in which the secondary transfer unit **40** is assembled to the drawer housing **113**. This facilitates the assembly of the secondary transfer unit **40** to the drawer housing **113**. The groove of the second support has an arc shape having a center on an axial center of the second shaft and an opening at an entrance of the groove in the assembly.

(69) FIG. 8 is a diagram of a rotary shaft **117** on the rear side according to a first modification of an embodiment of the present disclosure.

(70) In the present modification illustrated in FIG. 8, the rotary shaft **117** on the rear side includes an upper portion **123b** that is placed upstream from the groove **117a** of the rotary shaft **117** on the rear side in the assembly direction of the secondary transfer unit **40**. The upper portion **123b** has a guide face **121** that guides the rear end portion of the shaft **118**. The guide face **121** is formed by

cutting out an area of the upper portion **123b** that is adjacent to the shaft insertion port **119**, obliquely to the right with respect to the Z-direction in FIG. **8**.

(71) As illustrated in FIG. **8**, by incorporating the guide face **121**, when the rear end of the shaft **118** is offset to the left (or in the +Y-direction) during the assembly of the secondary transfer unit **40**, the shaft **118** comes into contact with the guide face **121**. Then, the shaft **118** is guided to the right (or in the -Y-direction) in FIG. **8** by the guide face **121**, and moves to the shaft insertion port **119** of the groove **117a**. Thus, the rear end portion of the shaft **118** can be inserted into the groove **117a** through the shaft insertion port **119**. This facilitates the assembly of the secondary transfer unit **40** to the drawer housing **113**.

(72) Further, as illustrated in FIG. **8**, forming a notch to expose the lower surface **122a** of the groove **117a**, when viewed from above, provides the following advantages. In other words, when the shaft **118** guided by the guide face **121** reaches a joint A between the guide face **121** and the shaft insertion port **119**, and contact with the guide face **121** is released, the rear end portion of the shaft **118** moves downward. When viewed from above, a notch is formed to expose the lower surface **122a** of groove **117a**. As a result, when the contact with guide face **121** is released and the rear end portion of shaft **118** moves downward, the shaft **118** comes into contact with the lower surface of groove **117a**, allowing the shaft **118** to be inserted into the groove **117a**.

(73) The tilt angle θ of the guide face **121** relative to the Z-direction, which is the assembly direction, is preferably 45 degrees or less. By setting the tilt angle of the guide face **121** to 45 degrees or less, when the rear end portion of the shaft **118** comes into contact with the guide face **121**, the shaft **118** can be smoothly moved along the guide face under the weight of the secondary transfer unit. This facilitates the insertion of the rear end portion of the shaft **118** into the groove **117a**, making it easier to assembly the secondary transfer unit **40** into the drawer housing **113**.

(74) As illustrated in FIG. **8**, the upper portion **123b** of the rotary shaft **117** on the rear side is notched in an area adjacent to the shaft insertion port **119** (the right side in FIG. **8**). This notch ensures that the length of the shaft insertion port **119** in the Y-direction is longer than the diameter **L1** of the shaft **118**. In this configuration, the length of the shaft insertion port **119** in the Y-direction is longer than the diameter **L1** of the shaft **118**. This configuration facilitates the insertion of the shaft **118** into the shaft insertion port **119** more than a configuration that the length of the shaft insertion port **119** is shorter than the diameter **L1** of the shaft **118**. This facilitates the assembly of the secondary transfer unit **40** to the drawer housing **113**.

(75) FIGS. **9A** and **9B** are diagrams each illustrating a rotary shaft **117** on the rear side according to a second modification of an embodiment of the present disclosure.

(76) In the second modification as illustrated in FIGS. **9A** and **9B**, a length **d1** between a joint C and a joint B in the Y-direction is made longer than the diameter **L1** of the shaft **118**. The joint C is between the guide face **121** and the outer circumferential surface **120**, and the joint B is between the shaft insertion port **119** and the outer circumferential surface **120** of the rotary shaft **117** on the rear side.

(77) In FIG. **9A**, similarly to FIG. **8**, the upper portion **123b** has a notch in the area adjacent to the shaft insertion port **119**. Further, the upper portion **123b** has a guide face **121**. In FIG. **9B**, the lower portion **123a** that is downstream from the groove **117a** of the rotary shaft **117** on the rear side in the assembly direction has a notch in an area adjacent to the shaft insertion port. The lower portion **123a** has a guide face **121**. In FIGS. **9A** and **9B**, the length of the shaft insertion port **119** in the Y-direction is shorter than the diameter **L1** of the shaft **118**.

(78) However, a length **d1** between a joint C and a joint B (in FIG. **9A**) or a joint A (in FIG. **9B**) in the Y-direction is longer than the diameter **L1** of the shaft **118**. The joint C is between the guide face **121** and the outer circumferential surface **120**, and the joint B or A is between the shaft insertion port **119** and the outer circumferential surface **120** of the rotary shaft **117** on the rear side.

Compared to a case where the length **d1** in the Y-direction is shorter than the diameter **L1** of the shaft **118**, the configuration of FIGS. **9A** and **9B** makes it easier for the shaft **118** to contact the

guide face **121** when the rear side of the shaft **118** shifts in the Y-direction during the assembly of the secondary transfer unit **40**. This facilitates the insertion of the rear end portion of the shaft **118** into the groove **117a**, making it easier to assembly the secondary transfer unit **40** into the drawer housing **113**.

(79) Additionally, incorporating the guide face **121** facilitates the insertion of the rear end portion of the shaft **118** into the groove **117a** even when the length of the shaft insertion port **119** in the Y-direction is shorter than the diameter **L1** of the shaft **118**. In FIG. **9A**, after being guided by the guide face **121**, the shaft **118** comes into contact with the lower surface of the groove **117a**, which is exposed from the shaft insertion port **119**, when viewed from above. Subsequently, the shaft is guided downward to the left in the figure by this lower surface, allowing it to be inserted into the groove **117a**.

(80) However, in FIG. **9B**, the rear end portion of the shaft **118** is guided downward to the left in the figure by the guide face **121**, allowing it to be inserted into the groove **117a**.

(81) In the support unit, the second support includes a first portion upstream from the groove in the assembly direction; and a second portion downstream from the groove in the assembly direction. At least one of the first portion or the second portion has the guide face, and the second support has an entrance distance between a first upstream end of the guide face or the opening of the first portion and a second upstream end of the guide face or the opening of the second portion in the orthogonal direction, equal to or longer than a diameter of the first shaft.

(82) The entrance distance is longer than a length of the shaft insertion port in the orthogonal direction.

(83) FIG. **10** is a diagram of a rotary shaft **117** on the rear side according to a third modification of an embodiment of the present disclosure.

(84) In the third modification as illustrated in FIG. **10**, each of the upper portion **123b** and the lower portion **123a** of the rotary shaft **117** has a guide face **121**.

(85) By incorporating the guide faces **121** on both the upper portion **123b** and the lower portion **123a**, the rear end portion of the shaft **118** can be guided by the guide face **121** regardless of whether it shifts to the left (or in the +Y-direction) or to the right (or in the -Y-direction) during the assembly of the secondary transfer unit **40**. This facilitates the insertion of the shaft **118** into the groove **117a**.

(86) In the third modification as illustrated in FIG. **10**, a length **d2** in the Y-direction between a joint **C1** and a joint **C2** is preferably equal to or longer than the diameter **L1** of the shaft **118**. The joint **C1** is between the guide face of the upper portion **123b** of the rotary shaft **117** and the outer circumferential surface of the rotary shaft **117**. The joint **C2** is between the guide face of the lower portion **123a** of the rotary shaft **117** and the outer circumferential surface of the rotary shaft **117**. Compared to a case where the length **d2** in the Y-direction is shorter than the diameter **L1** of the shaft **118**, the configuration of FIG. **10** makes it easier for the shaft **118** to contact the guide face **121** when the rear side of the shaft **118** shifts in the Y-direction during the assembly of the secondary transfer unit **40**. This facilitates the insertion of the rear end portion of the shaft **118** into the groove **117a**, making it easier to assembly the secondary transfer unit **40** into the drawer housing **113**.

(87) In the third modification, as illustrated in FIG. **11A**, the length of the shaft insertion port **119** in the Y-direction is preferably longer than the diameter **L1** of the shaft **118**. As illustrated in FIGS. **11B** and **11C**, a length **d3** between a joint **C2** (in FIG. **11B**) or a joint **C1** (in FIG. **11C**) and a joint **A** (in FIG. **11B**) or a joint **B** (in FIG. **11C**) in the Y-direction is longer than the diameter **L1** of the shaft **118**. The joint **C2** (in FIG. **11B**) or **C1** (in FIG. **11C**) is between the guide face **121** and the outer circumferential surface **120**, and the joint **A** (in FIG. **11B**) or **B** (in FIG. **11C**) is between the shaft insertion port **119** and the outer circumferential surface **120** of the rotary shaft **117** on the rear side. In the third modification as illustrated in FIGS. **11A** to **11C**, the rear end portion of the shaft **118** can be inserted into the groove **117a** more easily than the configuration of FIG. **10**.

(88) In the support unit, the second support includes: a first portion upstream from the groove in the assembly direction; a second portion downstream from the groove in the assembly direction; and multiple guide faces including the guide face. The multiple guide faces includes: a first guide face on a first upstream end of the first portion; and a second guide face on a second upstream end of the second portion. The second support has an entrance distance between the first upstream end of the first portion and the second upstream end of the second portion, equal to or longer than a diameter of the first shaft in the orthogonal direction.

(89) The entrance distance between the first upstream end and the second upstream end of the second portion is longer than a length of the shaft insertion port in the orthogonal direction.

(90) FIG. 12 is a diagram of a rotary shaft 117 on the rear side according to a fourth modification of an embodiment of the present disclosure.

(91) In the fourth modification as illustrated in FIG. 12, the width of the groove gradually increases in a direction toward the shaft insertion port 119. Additionally, the length of the shaft insertion port 119 in the Y-direction is equal to or longer than the diameter L1 of the shaft 118. With this configuration, the insertion of the shaft 118 into the shaft insertion port 119 from above is facilitated compared to cases where the length of the shaft insertion port 119 in the Y-direction is less than the diameter L1 of the shaft 118. Thus, the shaft 118 can be easily inserted into the groove 117a through the shaft insertion port 119 during the assembly of the second transfer unit to the drawer housing.

(92) As described with reference to FIGS. 4A-1 to 6B-2, when the secondary transfer roller 41 is pressed against the intermediate transfer belt 31 and the secondary transfer unit rotates around the Y-axis, a portion of the groove outside the intended movement range of the shaft 118 gradually expands. For example, by reducing the radius of curvature on at least one of the lower surface and the upper surface of the groove outside the intended movement range compared to within that movement range, the width of the groove outside this range can be gradually increased.

(93) As illustrated in FIGS. 8 to 11 in which the rotary shaft 117 on the rear side is notched to provide one or more guide faces 121, the joint between the guide face 121 and the shaft insertion port 119 is angled. The angular shape of the joint may cause the shaft 118 to get snagged and potentially damage it during attachment or detachment from the groove 117a. As illustrated in FIG. 12, the configuration lacks any angular portions, which prevents damage to the shaft 118 during attachment or detachment from the groove 117a.

(94) The above-described embodiments are illustrative and do not limit the present disclosure. Thus, numerous additional modifications and variations are possible in light of the above teachings. For example, elements and/or features of different illustrative embodiments may be combined with each other and/or substituted for each other within the scope of the present disclosure.

(95) In the description above, the present embodiment is applied to a drawer-type support unit that holds the secondary transfer unit as the pressure unit. However, the present embodiment can also be applied to other support units, such as those supporting the pressure rollers in fixing devices, for example.

(96) The configurations according to the above-described embodiments are examples, and embodiments of the present disclosure are not limited to the above. For example, the following aspects can achieve effects described below.

(97) Aspect 1

(98) A support unit includes a first support (e.g., the rotary shaft 116 on the front side) to support and position one end of a first shaft (or the reference shaft 118) of a first shaft of a pressure unit (e.g., the secondary transfer unit 40) including a pressure rotator (e.g., the secondary transfer roller 41) to apply pressure to an object (e.g., the intermediate transfer belt 31), in each of: an assembly direction (e.g., the Z-direction) in which the pressure unit is assembled to the support unit; and an orthogonal direction (e.g., the Y-direction) orthogonal to each of the assembly direction and an axial direction (e.g., the X-direction) of the pressure unit; a second support (e.g., the rotary shaft

117 on the rear side) having a groove (e.g., the groove **117a**) to support another end of the first shaft; and a third support (e.g., the contact surfaces **110** and **111**) supporting a second shaft (e.g., the secondary transfer roller shaft **106**) of the pressure rotator. The first support, the second support, and the third support the pressure unit to be rotatable around the reference shaft. The groove has a shape of an arc having a center on an axial center of the second shaft; and an opening as viewed in the assembly direction.

(99) Due to manufacturing errors, when the pressure rotator contacts the pressed member (or a pressurized object), the distance between one side of the pressure rotator and the pressurized object may differ from the distance between its opposite side and the pressurized object along the axis. When positioning the pressure unit, if both ends of the reference shaft are aligned both with the direction of assembly into the support unit and perpendicular to both the assembly direction and the axis direction, the pressure unit may become twisted.

(100) Specifically, after the closer side of the pressure member or the pressure rotator contacts the pressurized object, the pressure unit may twist. This twisting action accommodates the difference in distance between one end (one side) of the pressure rotator and the another end (its opposite side) relative to the pressurized object. Eventually, this allows the side of the pressure rotator that was initially further away to also make contact with the pressurized object. Thus, if twisting occurs in the pressure unit, it could disrupt the positional relationships between components installed within the pressure unit.

(101) In contrast, in Aspect 1, the another end of the reference shaft is supported by the groove portion of the second support portion so as to be movable in an arc shape around the axial center of the shaft of the pressure member. As a result, after the end (closer side) of the pressure rotator that is closer to the pressurized object than the another end along the axial direction contacts the pressurized object, the another end (distal side) is brought into contact with the pressurized object. As the another end of the reference shaft moves within a groove, the pressure unit tilts with respect to the axial direction when viewed in the orthogonal direction. This compensates for any differences in distance between the two ends of the pressure rotator due to manufacturing errors or other factors, enabling the distal side of the pressure rotator to contact the pressurized object.

(102) By tilting the pressure unit, it can absorb the differences in distance between the ends of the pressure rotator and the pressurized object. This helps to prevent twisting in the pressure unit when the pressure rotator applies pressure to the pressurized object.

(103) In Aspect 1, the groove is shaped such that its opening is visible when viewed in the direction of assembly. This allows the another end of the reference shaft to be easily inserted into the groove by simply moving the pressure unit in the direction of assembly toward the support unit. Thus, the assembly of the pressure unit to the support unit is facilitated.

(104) Aspect 2

(105) In the aspect 1, when viewed in the axial direction (e.g., the X-direction), the opening (e.g., the shaft insertion port **119**) has a length from one end to the another end in the orthogonal direction (e.g., Y-direction) is longer than a diameter of the reference shaft (e.g., the shaft **118**).

(106) This configuration allows the reference shaft such as the shaft **118** to be easily inserted into the opening such as the shaft insertion port **119** in the assembly direction (Z-direction) and also facilitates the assembly of the pressure unit such as the secondary transfer unit **40** to the support unit such as the drawer housing **113**.

(107) Aspect 3

(108) In the aspect 2, the second support (e.g., the rotary shaft **117** on the rear side) includes: a first portion (e.g., the upper portion **123b**) upstream from the groove (e.g., the groove **117a**) in the assembly direction; and a second portion (e.g., the lower portion **123a**) downstream from the groove in the assembly direction. At least one of the first portion or the second portion has a notch adjacent to the opening.

(109) With this configuration, as described in the above embodiment, the length between one end

and the another end of the opening (e.g., the shaft insertion port **119**) in the orthogonal direction (e.g., the Y-direction) as viewed in the axial direction (e.g., the X-direction) can be made longer than the diameter of the reference shaft such as the shaft **118**.

(110) Aspect 4

(111) In Aspect 2, a width of the groove (e.g., the groove **117a**) gradually increases toward the opening (e.g., the shaft insertion port **119**).

(112) With this configuration, as described with reference to FIG. **12** above, the length between one end and the another end of the opening (e.g., the shaft insertion port **119**) of the groove in the orthogonal direction (e.g., the Y-direction) as viewed in the axial direction (e.g., the X-direction) can be made longer than the diameter of the reference shaft such as the shaft **118**.

(113) Compared to a case where the length of the reference-shaft insertion port (e.g., the shaft insertion port **119**) from one end to the another end in the orthogonal direction (e.g., the Y-direction) is longer than the diameter of the reference shaft (e.g., the shaft **118**) when viewed in the axial direction (e.g., the X-direction), the notch prevents the shaft **118** from getting snagged and prevents damage to the shaft **118** during attachment or detachment from the groove **117a**.

(114) Aspect 5

(115) In any one of Aspects 1 to 4, the opening includes a reference-shaft insertion port through which the another end of the reference shaft is inserted into the groove (e.g., the groove **117a**). The second support (e.g., the rotary shaft **117** on the rear side) has a guide face **121** tilted relative to the assembly direction (or the groove) to guide the another end of the reference shaft (e.g., the shaft **118**) to the reference-shaft insertion port (e.g., the shaft insertion port **119**).

(116) With this configuration, as described in the above embodiment, when the reference shaft is offset from the reference-shaft insertion port (e.g., the shaft insertion port **119**) in the vertical direction (e.g., the Y-direction) and contacts the guide face **121**, the reference shaft (e.g., the shaft **118**) of the secondary transfer unit **40** is guided by the guide face **121** into the reference-shaft insertion port. Thus, the reference shaft can be easily inserted into the reference-shaft insertion port, and the pressure unit (e.g., the secondary transfer unit **40**) can be easily assembled to the support unit (e.g., the drawer housing **113**).

(117) Aspect 6

(118) A support unit includes a first support (e.g., the rotary shaft **116** on the front side) to support and restrict one end of a first shaft (or the reference shaft **118**) in each of: an assembly direction (e.g., the Z-direction) in which the pressure unit is assembled to the support unit; and an orthogonal direction (e.g., the Y-direction) perpendicular to each of the assembly direction and an axial direction (e.g., the X-direction) of the pressure unit, the first shaft to position a pressure unit (e.g., the secondary transfer unit **40**) including a pressure rotator (e.g., the secondary transfer roller **41**) to pressurize a pressurized object (e.g., the intermediate transfer belt **31**); a second support (e.g., the rotary shaft **117** on the rear side); and a third support (e.g., the contact surfaces **110** and **111**) supporting a second shaft (e.g., the secondary transfer roller shaft **106**) of the pressure rotator. The second support has a groove (e.g., the groove **117a**) to support the another end of the first shaft; a shaft insertion port (e.g., the shaft insertion port **119**) through which the another end of the first shaft is inserted into the groove (e.g., the groove **117a**); and a guide face (e.g., the guide face **121**) tilted relative to the assembly direction to guide the another end of the first shaft to the shaft insertion port. The first support, the second support, and the third support supports the pressure unit to be rotatable around the reference shaft. The groove has a shape of an arc centered on a center of the second shaft (e.g., the shaft of the pressure rotator).

(119) Similarly to Aspect 1, the another end of the reference shaft is supported by the groove of the second support so as to be movable in an arc shape around the center of the shaft of the pressure rotator such as the secondary transfer roller **41**.

(120) As a result, after the end (closer side) of the pressure rotator that is closer to the pressurized object than the another end along the axial direction contacts the pressurized object, the another end

(distal side) is brought into contact with the pressurized object. As the another end of the reference shaft moves within a groove, the pressure unit tilts with respect to the axial direction when viewed in the orthogonal direction (e.g., the Y-direction). As it tilts, the another end (the distal end) of the pressure rotator that is away from the pressure unit approaches the pressurized object (e.g., the intermediate transfer belt **31**), allowing this end (the distal end) of the pressure rotator to make contact with the pressurized object. Thus, the pressure unit may twist when the pressing member applies pressure to the pressurized object.

(121) With this configuration, as described in the above embodiment, when the reference shaft is offset from the reference-shaft insertion port (e.g., the shaft insertion port **119**) in the vertical direction (e.g., the Y-direction) and contacts the guide face **121**, the reference shaft (e.g., the shaft **118**) of the secondary transfer unit **40** is guided by the guide face **121** into the reference-shaft insertion port. Thus, the reference shaft can be easily inserted into the reference-shaft insertion port, and the pressure unit (e.g., the secondary transfer unit **40**) can be easily assembled to the support unit (e.g., the drawer housing **113**).

(122) Aspect 7

(123) In Aspect 5 or 6, the second support (e.g., the rotary shaft **117** on the rear side) includes: a first portion (e.g., the upper portion **123b**) upstream from the groove (e.g., the groove **117a**) in the assembly direction; and a second portion (e.g., a portion **123a**) downstream from the groove in the assembly direction. At least one of the first portion or the second portion has the guide face (e.g., the guide face **121**). In a plane orthogonal to the axial direction or when viewed in the axial direction, a distance (e.g., the length $d1$) between a first point (e.g., a joint A) and a second point in the orthogonal direction is equal to or longer than a diameter $L1$ of the first shaft (e.g., the reference shaft). The first point is a joint between the guide face and an outer circumferential surface **120** of one of the first portion and the second portion, and the second point is an edge of the shaft insertion port (e.g., the shaft insertion port **119**) that is adjacent to the another one of the first portion and the second portion.

(124) According to this configuration, as described with reference to FIG. **9**, the reference shaft (e.g., the shaft **118**) is more likely to contact the guide face **121** when the another end of the reference shaft shifts in the orthogonal direction (e.g., the Y-direction) during the assembly of the pressure unit (e.g., the secondary transfer unit **40**), compared to the case where the length $d1$ is less than the diameter $L1$ of the reference shaft. Thus, the reference shaft can be further easily inserted into the reference-shaft insertion port, and the pressure unit (e.g., the secondary transfer unit **40**) can be easily assembled to the support unit (e.g., the drawer housing **113**).

(125) Aspect 8

(126) In Aspect 7, in the plane perpendicular to the axial direction (or when viewed in the axial direction (e.g., in the Y-direction), the distance between the first point (e.g., the joint A) and the second point in the orthogonal direction is longer than a length of the shaft insertion port (e.g., the shaft insertion port **119**) in the orthogonal direction. The first point is a joint (e.g., the joint A) between the guide face (e.g., the guide face **121**) and an outer circumferential surface **120** of one of the first portion (e.g., the upper portion **123b** upstream from the groove in the assembly direction) and the second portion (e.g., the lower portion **123a** downstream from the groove in the assembly direction), and the second point is an edge of the shaft insertion port (e.g., the shaft insertion port **119**) that is adjacent to the another one of the first portion and the second portion.

(127) With this configuration, even when the length of the reference-shaft insertion opening in the orthogonal direction is shorter than the diameter of the reference shaft, the reference shaft can be guided by the guide face and the reference shaft can be easily inserted into the reference-shaft insertion opening.

(128) Aspect 9

(129) In Aspect 5 or 6, the support unit further includes multiple guide faces (e.g., the guide faces **121**) including the guide face (e.g., the guide face **121**). The second support (e.g., the rotary shaft

117 on the rear side) includes: a first portion (e.g., the upper portion **123b**) upstream from the groove (e.g., the groove **117a**) in the assembly direction (e.g., the Z-direction) and having a first guide face (e.g., the guide face **121**) of the multiple guide faces; and a second portion (e.g., the lower portion **123a**) downstream from the groove in the assembly direction and having a second guide face (e.g., the guide face **121**) of the multiple guide faces. In a plane orthogonal to the axial direction (when viewed in the axial direction, e.g., the Z-direction), a distance (the length **d2**) between a first joint (e.g., the joint **C1**) and a second joint (the joint **C2**) in the orthogonal direction is equal to or longer than a diameter (e.g., the diameter **L1**) of the first shaft (e.g., the reference shaft, the shaft **118**) where the first joint is between the first guide face and a first outer circumferential surface **120** of the first portion, and the second joint is between the second guide face and a second outer circumferential surface **120** of the second portion.

(130) According to this configuration, as described with reference to FIG. **10**, the reference shaft (e.g., the shaft **118**) is more likely to contact the guide face **121** when the another end of the reference shaft shifts in the orthogonal direction (e.g., the Y-direction) during the assembly of the pressure unit (e.g., the secondary transfer unit **40**), compared to the case where the length **d1** is less than the diameter **L1** of the reference shaft. Thus, the reference shaft can be further easily inserted into the reference-shaft insertion port, and the pressure unit (e.g., the secondary transfer unit **40**) can be easily assembled to the support unit (e.g., the drawer housing **113**).

(131) Aspect 10

(132) In the aspect 9, in the plane orthogonal to the axial direction (when viewed in the axial direction, e.g., the Y-direction), the distance between the first joint (e.g., the joint **C1**) and the second joint (e.g., the joint **C2**) in the orthogonal direction (e.g., in the Y-direction) is longer than a length of the shaft insertion port in the orthogonal direction.

(133) With this configuration, even when the length of the reference-shaft insertion opening in the orthogonal direction is shorter than the diameter of the reference shaft, the reference shaft can be guided by the guide face and the reference shaft can be easily inserted into the reference-shaft insertion opening.

(134) Aspect 11

(135) In any one of Aspects 5 to 10, a tilt angle of the guide face (e.g., the guide face **121**) relative to the assembly direction (e.g., the Z-direction) is 45 degrees or less.

(136) With this configuration, as described in the above embodiment, when the pressure unit (e.g., the secondary transfer unit **40**) is assembled to the support unit (e.g., the drawer housing **113**), the reference shaft can be moved relative to the guide face by a weak force in the assembly direction. Thus, the reference shaft can be easily inserted into the reference-shaft insertion port.

(137) Aspect 12

(138) In any one of Aspects 1 to 11, the support unit further includes a pressing body (e.g., the rotary arms **108** and **109**) including the third support (e.g., the contact surfaces **110** and **111**) that supports the second shaft (e.g., the secondary transfer roller shaft **106**) of the pressure rotator, to press the pressure rotator against the pressurized object.

(139) According to this configuration, the third support presses the shaft (e.g., the secondary transfer roller shaft **106**) of the pressure rotator (e.g., the secondary transfer roller **41**) so as to press the pressure rotator (e.g., the secondary transfer roller **41**) against the pressurized object (e.g., the intermediate transfer belt).

(140) Aspect 13

(141) A transfer device includes multiple tension rollers including a transfer roller (e.g., the secondary transfer roller **41**) as the pressure rotator; and a transfer belt (e.g., the secondary transfer belt **102**) looped around the multiple tension rollers; a pressure unit (e.g., the secondary transfer unit **40**) to press the transfer roller against the pressurized object including an image carrier (e.g., the intermediate transfer belt **31**) via the transfer belt to form a transfer nip; and the support unit (e.g., the drawer housing **113**) according to any one of Aspects 1 to 12 supporting the pressure unit.

(142) With this configuration, the occurrence of twisting in the pressure unit (e.g., the secondary transfer unit) is reduced, and the pressure unit can be easily assembled to the support unit (e.g., the drawer housing **113**).

(143) Aspect 14

(144) In an image forming apparatus including a transfer device, the transfer device according to Aspect 13 is used as the transfer device.

(145) With this configuration, the occurrence of twisting in the pressure unit (e.g., the secondary transfer unit) is reduced, and the pressure unit can be easily assembled to the support unit (e.g., the drawer housing **113**).

(146) The above-described embodiments are illustrative and do not limit the present invention. Thus, numerous additional modifications and variations are possible in light of the above teachings. For example, elements and/or features of different illustrative embodiments may be combined with each other and/or substituted for each other within the scope of the present invention.

Claims

1. A support unit comprising: a first support to support and position one end of a first shaft of a pressure unit, including a pressure rotator to apply pressure to an object, in each of: an assembly direction in which the pressure unit is assembled to the support unit; and an orthogonal direction orthogonal to each of the assembly direction and an axial direction of the first shaft; a second support having a groove supporting another end of the first shaft; and a third support supporting a second shaft of the pressure rotator, wherein the first support, the second support, and the third support the pressure unit rotatable around the first shaft, and the groove of the second support has: an arc shape having a center on an axial center of the second shaft; and an opening at an entrance of the groove in the assembly direction.
2. The support unit according to claim 1, wherein the opening has a length longer than a diameter of the first shaft in the orthogonal direction.
3. The support unit according to claim 2, wherein the second support includes: a first portion upstream from the groove in the assembly direction; and a second portion downstream from the groove in the assembly direction, and at least one of the first portion or the second portion has a notch adjacent to the opening.
4. The support unit according to claim 2, wherein a width of the groove gradually increases toward the opening in the assembly direction.
5. The support unit according to claim 1, wherein the opening includes a shaft insertion port through which the another end of the first shaft is inserted into the groove, and the second support has a guide face tilted relative to the groove to guide the another end of the first shaft to the shaft insertion port.
6. The support unit according to claim 5, wherein the second support includes: a first portion upstream from the groove in the assembly direction; and a second portion downstream from the groove in the assembly direction, at least one of the first portion or the second portion has the guide face, and the second support has an entrance distance, between a first upstream end of the guide face or the opening of the first portion and a second upstream end of the guide face or the opening of the second portion in the orthogonal direction, equal to or longer than a diameter of the first shaft.
7. The support unit according to claim 6, wherein the entrance distance is longer than a length of the shaft insertion port in the orthogonal direction.
8. The support unit according to claim 5, wherein the second support includes: a first portion upstream from the groove in the assembly direction; a second portion downstream from the groove in the assembly direction; and multiple guide faces including the guide face including: a first guide face on a first upstream end of the first portion; and a second guide face on a second upstream end

of the second portion, and the second support has an entrance distance, between the first upstream end of the first portion and the second upstream end of the second portion, equal to or longer than a diameter of the first shaft in the orthogonal direction.

9. The support unit according to claim 8, wherein the entrance distance between the first upstream end of the first portion and the second upstream end of the second portion is longer than a length of the shaft insertion port in the orthogonal direction.

10. The support unit according to claim 5, wherein a tilt angle of the guide face relative to the assembly direction is 45 degrees or less.

11. The support unit according to claim 1, further comprising a pressing body including the third support, to press the pressure rotator against the object.

12. A transfer device comprising: multiple tension rollers including a transfer roller; and a transfer belt looped around the multiple tension rollers; a pressure unit to press the transfer roller against an image carrier via the transfer belt to form a transfer nip; and the support unit according to claim 1 supporting the pressure unit.

13. An image forming apparatus comprising the transfer device according to claim 12.

14. A support unit comprising: a first support to support and position one end of a first shaft of a pressure unit, including a pressure rotator to apply pressure to an object, in each of: an assembly direction in which the pressure unit is assembled to the support unit; and an orthogonal direction orthogonal to each of the assembly direction and an axial direction of the first shaft; a second support having: a groove supporting another end of the first shaft; a shaft insertion port through which the another end of the first shaft is inserted into the groove; and a guide face tilted relative to the groove to guide the another end of the first shaft to the shaft insertion port; and a third support supporting a second shaft of the pressure rotator, wherein the first support, the second support, and the third support the pressure unit rotatable around the first shaft, and the groove of the second support has an arc shape having a center on an axial center of the second shaft.
